

Project Definition: User Experience in the 3D Pipeline

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1 Introduction

In my project I will build on players existing user experience of games by making an interesting concept for a mini golf game, taking inspiration from other golf games and theories around visual design and game design to focus on the art and design aspect. The output will be a vertical slice of a playable game using the made Blender assets to put together a scene which is visually pleasing.

2 Aims

My aim is to create a mini golf experience in Unreal Engine through using Blender to make modular kits for a mini golf course centred around different genres and themes. It will be a playable mini golf course game, prioritising user experience for the world. The technical challenge would be creating all of the objects together to fit the same theme and aesthetic throughout the levels and themes, whilst being visually pleasing and complex.

3 Objectives

Must

- I **must** create effective art and have justification for my design ideas.
- I **must** create 3 mini golf levels. This will be done using my modular kit and also by creating hero assets to be used in certain scenes to add to the themes in each part of the level.
- Each level **must** consist of 18 holes, as is typical for mini golf courses.

Should

- The levels **should** be challenging. It is a user experience in implementation to make this difficult for the user by having long 'holes' or twisted paths with obstructions trying to prevent the player from doing this easily.

Could

- The user **could** be able to play each level. This could be done by playing the ball to put the ball in the hole in the least number of strokes possible.

Won't

- The user **won't** be able to play this as a multiplayer game with their friends, as this is out of scope for the project.

- This **won't** be a game, as it does not include many mechanics and scoring systems resembling of a game, this is because this is out of scope. This project is a vertical slice of what a game could be.

4 Deliverables

Must

- I **must** focus on the art and blender assets.
- I **must** deliver a good 3 levels with this implementation, delivering a good visual experience to players.

Should

- I **should** focus on the scene within the level.
- I **should** implement the assets into Unreal Engine.

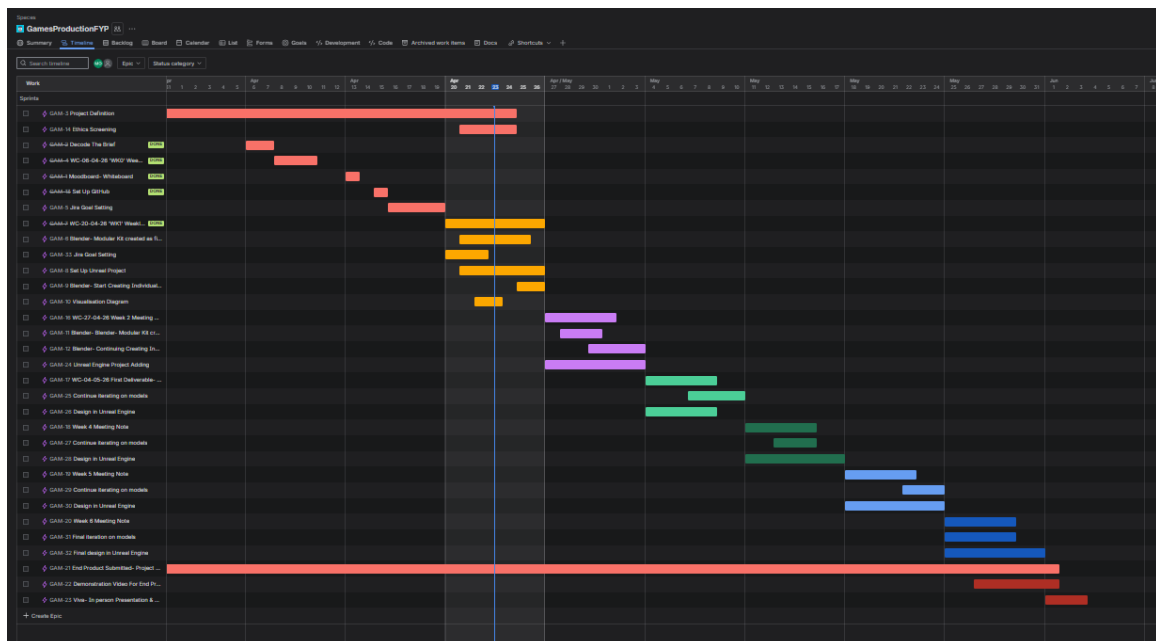
Could

- I **could** have functional golf gameplay.
- I **could** have UI only as a form to help this experience, including icons, or a scale measured to hit the ball with.

Won't

- I **won't** have a working pause menu with settings, as this is not what I'm showcasing in the project.
- This **won't** be a game, but is a vertical slice of the aspect of what a game could be in terms of art and design and to be visually pleasing.

5 Schedule of Activities



I will be using Sprint game development to organise the production of my product. This will make breaking down tasks into manageable objectives and clearer goals. This agile development philosophy is great for me to use as it will keep me on top of deadlines and ensures that I am always making progress with my product.

- **Sprint 0** - Heavily focused on pre-production and brainstorming product ideas. Understanding what is expected of me for this project and how I will accomplish this in the given time frame. In this weeks sprint I am setting out my objectives and goals that I wish to complete in each sprint and how to produce my project. This includes laying out my goals in the pre-production and production phases.
- **Sprint 1 - WC-20/04/2026** - Finish goal setting in Jira. - Visualisation diagrams, to start visualising what I'm going to make, keeping on track and on task within scope. - Making a start in Blender - Setting up Unreal Engine project.
- **Sprint 2 - WC-27/04/2026** - Hero assets, creating hero assets in Blender within theme. - Continue iterating on models. - Meeting notes must be up to date for submission. - Start designing in Unreal Engine
- **Sprint 3 - WC-04/05/2026** - First deliverable needs: GDD, TDD, and first prototype. - Week 3 meeting note. - Continue iterating on models. - Design in Unreal Engine.
- **Sprint 4 - WC-11/05/2026** - Week 4 Meeting note. - Continue iterating on models. - Design in Unreal Engine.
- **Sprint 5 - WC-18/05/2026** - Week 5 Meeting note. - Continue iterating on models. - Design in Unreal Engine.
- **Sprint 6 - WC-25/05/2026** - Week 6 Meeting note. - Final iteration and design of the models. - Final design in Unreal Engine.

6 Requirements

- **Blender 5.1-** Used to make 3D models and to aid in texturing and visual design.
- **Unreal Engine 5.5.4-** An interactable experience, Unreal Engine **should** be used to put all of the assets into the scene in a well thought out manner. The user **could** control a ball to progress in each level.
- **Adobe Creative Cloud-** I will be using Photoshop and other software to assist in visual aspects, such as texturing in Blender, and materials for objects (Unreal Engine).
- **Jira-** To organise myself and keep track of my projects direction.

7 State of the Art

- The assets could be collated together to make an asset pack for game developers to take from the asset store. The assets could be shown to the developers of 'Golf With Your Friends' (Team 17)[1], and they may decide to hire the person for making these assets as they could be seen as directly transferable to the game, for example in a DLC.
- Another game that I am drawing inspiration from is the game 'Golf It!'[2], which showcases it's modular kit by allowing the users to put together their own courses their own way however this won't be apparent in my game, it is good to see what other games have done in a similar art style to my project. It is worth noting that 'Golf It!' has a more high-poly models and detailed textures than 'Golf With Your Friends'.
- The modular kit could be seen as similar to the 'Kenney' game asset pack[3]. I **should** be proving the concept by putting the assets into Unreal Engine which **could** be with functionality.

8 Risk Factors

High Risk

- Avoiding mental blocks and tangents that do not contribute to the objectives of this assignment will be tricky, especially when trying to iteratively design the assets whilst implementing it in Unreal Engine.

Medium Risk

- Giving myself time to complete tasks and trying to stick to deadlines whilst taking into account unexpected circumstances such as falling ill will be challenging.

Low Risk

- Giving myself time to complete tasks to avoid unexpected errors in software disruption delaying progress, this is likely to happen if support for software is down but should only last for less than a day, usually due to a large cyberattack or necessary updates.

References

- [1] Golf With Your Friends. <https://store.steampowered.com/app/431240>.
- [2] Golf It!. <https://store.steampowered.com/app/571740/>.
- [3] Kenny. Minigolf kit by kenney (assets) <https://kenney-assets.itch.io/minigolf-kit>.

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